

Narratives of AI Innovation:

A Textual Analysis of Western Social Media's View on ChatGPT and DeepSeek

March 9th, 2025

Presentation Video Link: <https://www.youtube.com/watch?v=eMbICh3nIS8>

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Project Overview

Project Overview

Motivation

The rise of DeepSeek and other large language models (LLMs) marks both a technological milestone and an opportunity to explore the geopolitical and cultural tensions in the global AI landscape. This study examines how predominantly English-speaking social media platforms, particularly Reddit, shape perceptions of AI models like ChatGPT (developed in the U.S.) and DeepSeek (developed in China).

Research Question

What narratives emerge from Western Reddit communities' discussions about ChatGPT and DeepSeek, and how do these narratives reflect user perceptions of AI innovation?

Machine Learning Task Formulation

(1) **emotion/sentiment labeling** with RoBERTa-base-go_emotions (supervised), (2) **topic modeling** with BERTopic (unsupervised) to identify narrative themes, and (3) **clustering** (unsupervised) of BERTopic probability outputs to group similar narratives and answer the research question.

Theory of Orientalism

Edward Said's theory of Orientalism (1977) provides a critical framework for understanding how the West constructs the "Other," often framing Chinese AI technologies like DeepSeek as exotic, inferior, or threatening, while portraying Western models like ChatGPT as symbols of innovation and superiority. This study explores how Western social media users on Reddit (e.g., r/ChatGPT and r/DeepSeek) reinforce these Orientalist narratives, reflecting broader geopolitical tensions and cultural biases. While the Reddit community is predominantly tech-savvy and Western-centric, it offers valuable insights into how power dynamics shape global AI discourse, particularly in the context of rising competition between the West and China.

Hypothesis

- H1: Sentiment towards DeepSeek will be more negative and emotionally charged compared to ChatGPT, reflecting cultural stereotypes about Chinese LLMs. ‘
- H2: Discussions about Chinese LLMs will frequently feature terms like "surveillance" or "authoritarianism," while Western LLMs will emphasize "freedom," "creativity," and "leadership."
- H3: Clustering of DeepSeek discussions will group granular topics into broader themes, with "geopolitical risks" exhibiting more negative sentiment than "technical capabilities."

Data Processing & EDA

Data Processing & EDA

Data Preparation

Data Collection

- Source: [arctic-shift](#)
- Subreddits: *r/ChatGPT*, *r/DeepSeek*
- Size: 641, 961 in total
- Time Range: 2024/12/27 - 2025/03/03

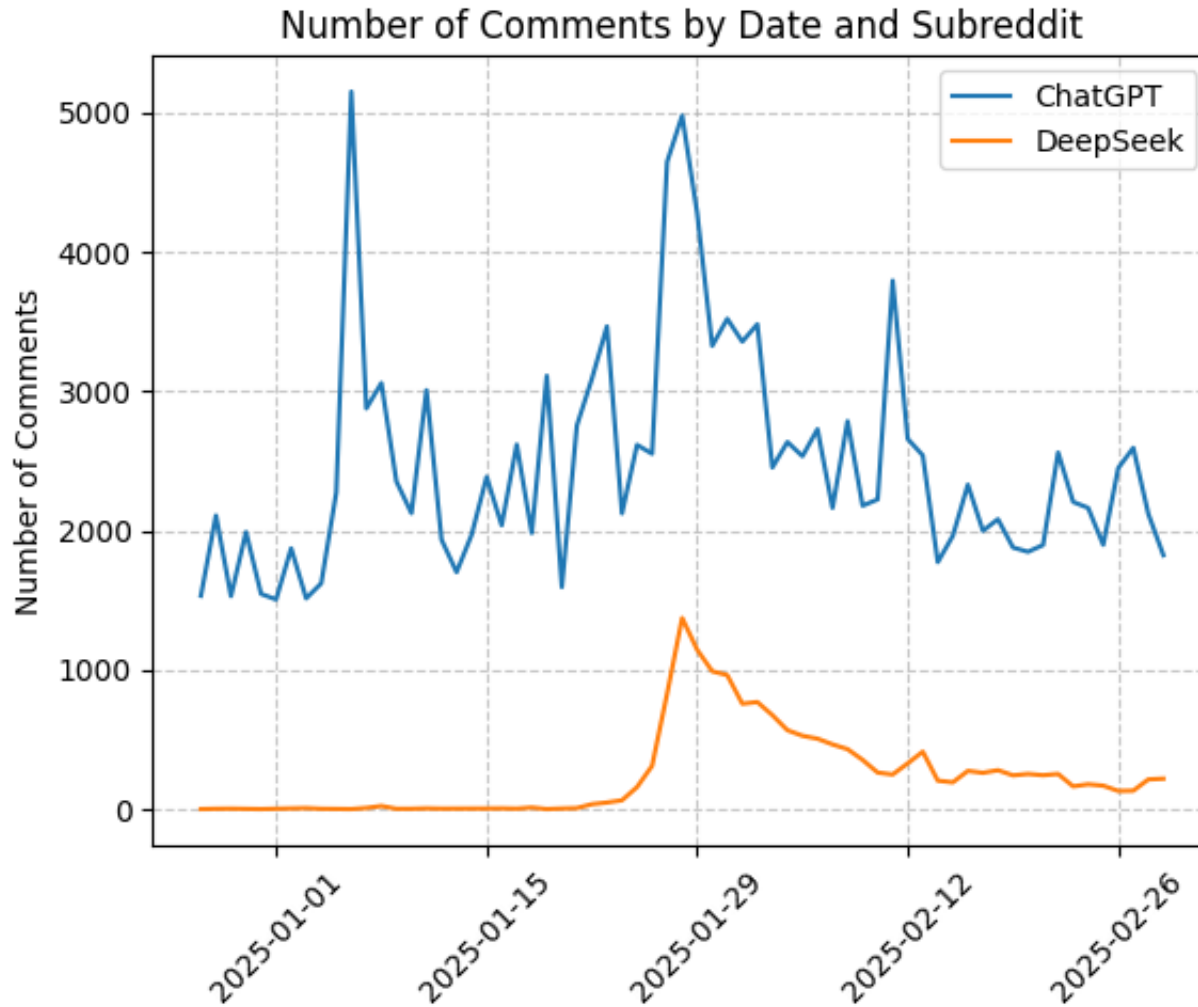
Preprocess data:

- Remove comments that below 12 words (medium quartile) and above 96 words (95th percentile)
- Remove mentions, special characters, URLs, bot replies
- Convert the unix timestamp to readable date

	subreddit	author	date	text	type
3	DeepSeek	Anuclano	2024-12-27	Those LLMs are notorious to not know who they ...	comments
7	DeepSeek	Positive_Average_446	2024-12-28	Doesn't its system prompt states its name thou...	comments
8	DeepSeek	Positive_Average_446	2024-12-28	I got this error when trying to log in with br...	comments
11	DeepSeek	Positive_Average_446	2024-12-28	Ah yep just tested its prompt only say AI assi...	comments
14	DeepSeek	redswan_cosignitor	2024-12-29	I actually noticed that it falls back on using...	comments
...	...	...	...	...	...
641951	ChatGPT	SeaBearsFoam	2025-03-01	This is the way I fed it chapters of my book t...	comments
641952	ChatGPT	headcanonball	2025-03-01	It's new and people haven't developed the anti...	comments
641954	ChatGPT	Postmanpale	2025-03-01	wants other humans to interact with his prints...	comments
641956	ChatGPT	LiteSoul	2025-03-01	Yeah I always go back to the extreme growth an...	comments
641958	ChatGPT	ThenExtension9196	2025-03-01	I think the bar is going to be raising We thin...	comments

253303 rows x 5 columns

Exploratory Data Analysis



- Important dates:
 - 2024/12/26: DeepSeek-V3 release date
 - 2024/12/27: First comment appear on r/DeepSeek
 - 2025/1/20: DeepSeek-R1 release date

Model Exploration/Interpretation

Emotional Narratives

Model: *SamLowe/roberta-base-go_emotions*

- *Widely used and trained based on Reddit data with 28 emotion labels*
- *Evaluation: Randomly sampled 50 input to check its label emotion*

Data Processing:

- Drop emotion category with less than 5 posts
- Drop neutral emotions
- Map emotions into sentiment of 'positive', 'negative', 'neutral, and 'ambiguous'

Topic Modeling

Model: *BERTopic*

- Traditional clustering methods (like K-Means) rely on Euclidean distance, which does not capture semantic similarity well.
- BERTopic converts documents into embeddings using transformer models (e.g., sentence-transformers), which better represent meaning.

Model Tuning:

- Set stop words
- Start without setting number of topics (too many identified topics and too large to run)
 - Set number of topics to 50, and reduced to 20, rearrange outliers
 - Stratified by date and random sampling from r/ChatGPT (1/10) to benchmark number of r/DeepSeek

Model Evaluation:

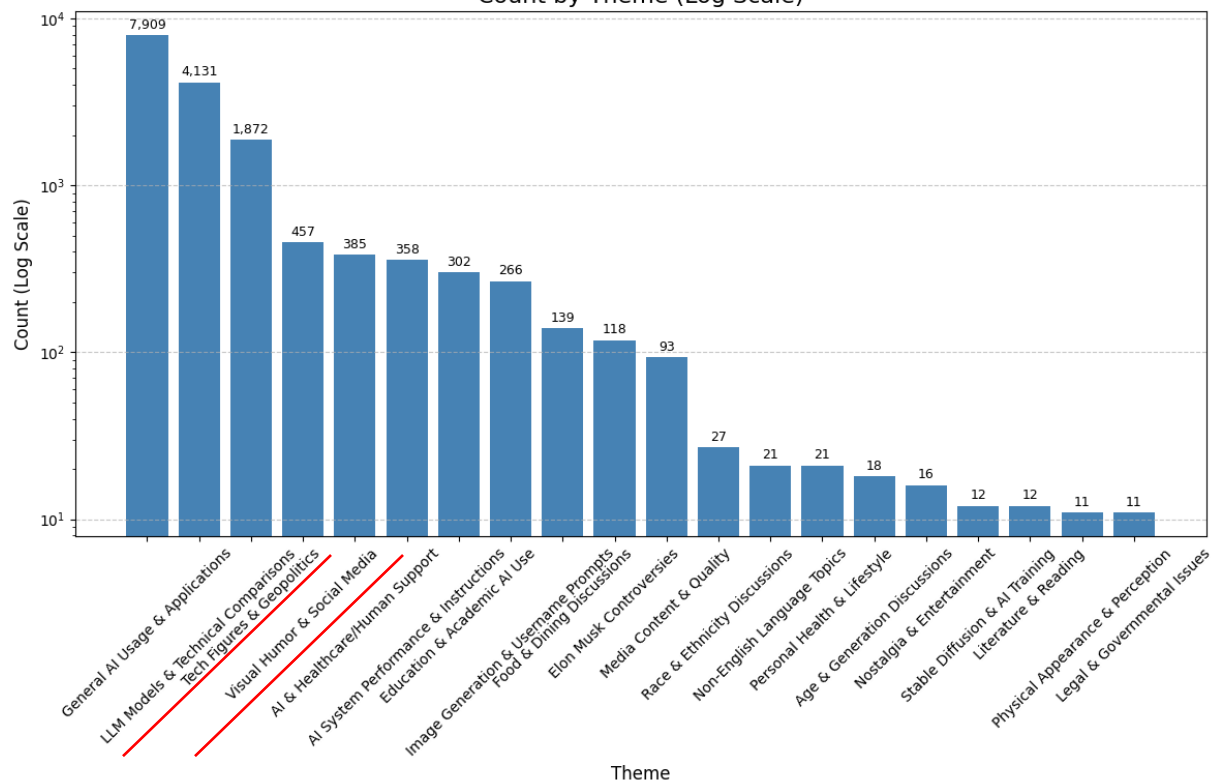
- Using LLM to assign themes
- Topic-document distribution
- Topic Coherence Score
- Topic Diversity

Topic Modeling - Model Evaluation

Topic-document distribution

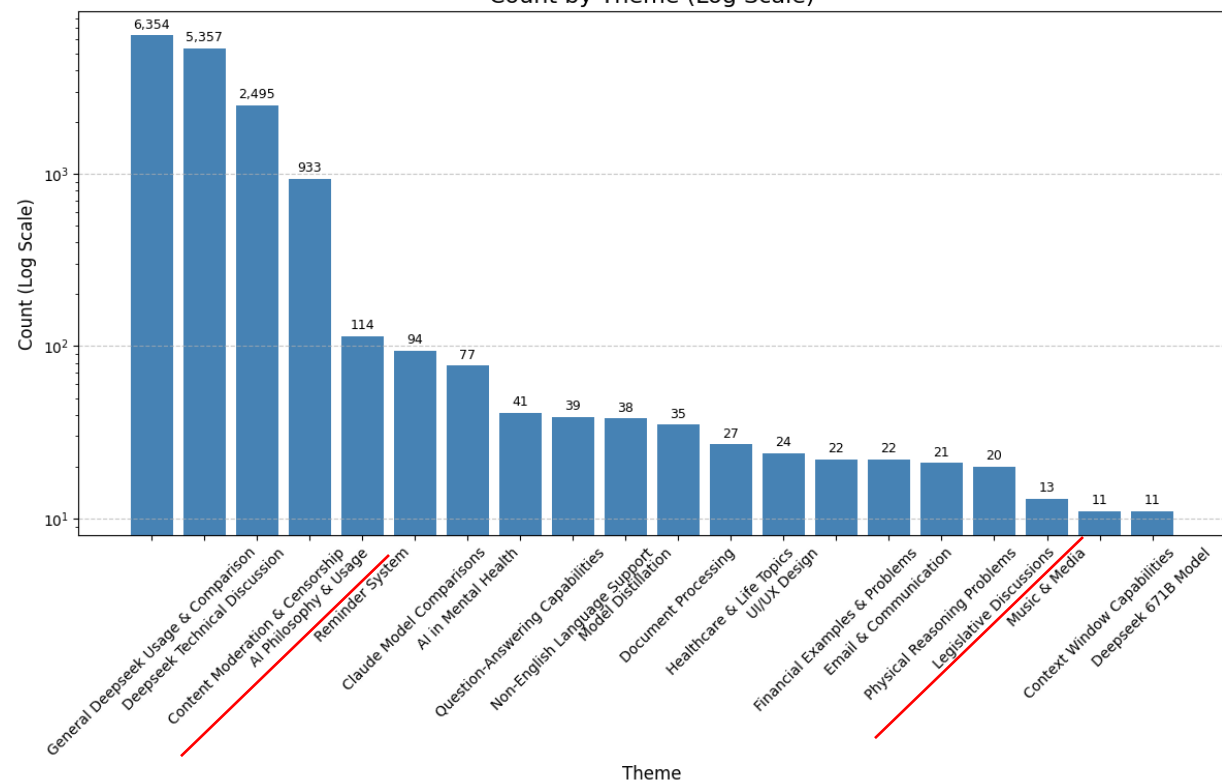
ChatGPT

Count by Theme (Log Scale)



DeepSeek

Count by Theme (Log Scale)



Topic Modeling - Model Evaluation

Topic Coherence Score & Topic Diversity

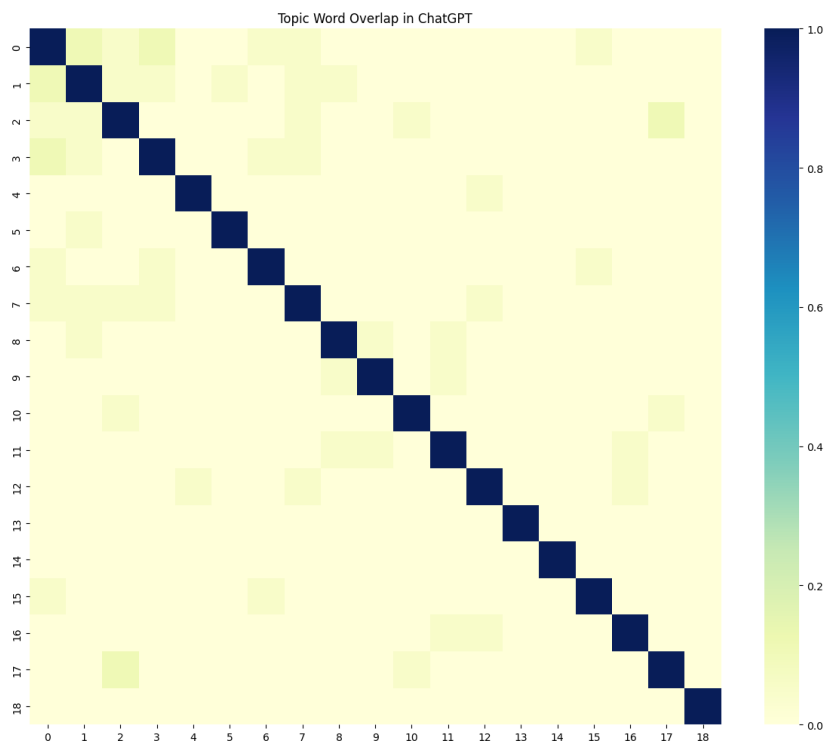
ChatGPT

Average topic overlap: 0.0087 (lower is better)

'num_topics': 19,

'diversity': 0.8842105263157894,

'avg_overlap': 0.008720631989330044



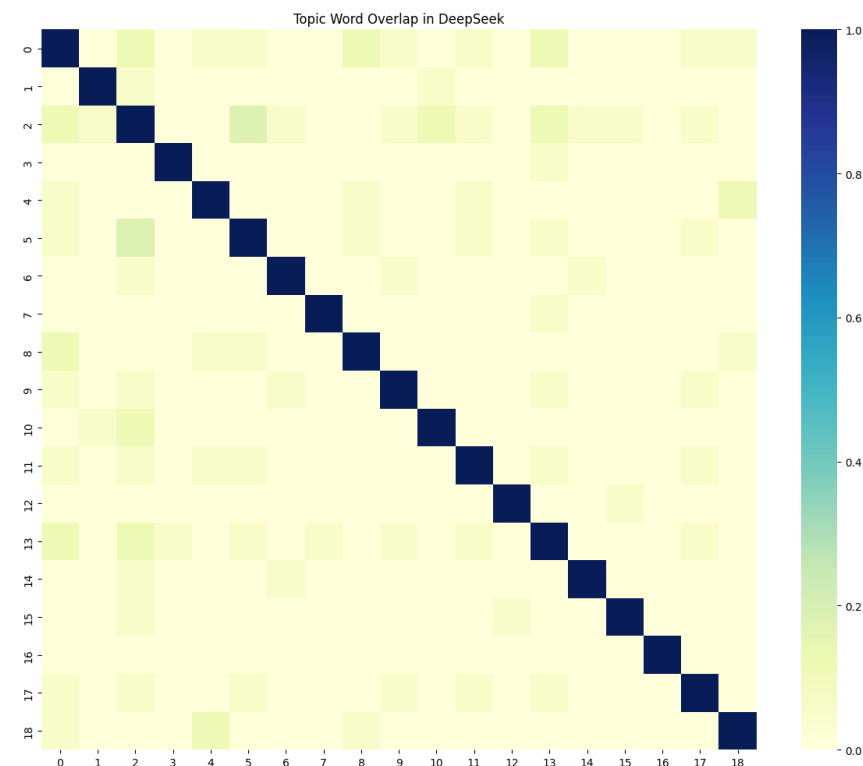
DeepSeek

Average topic overlap: 0.0145 (lower is better)

{'num_topics': 19,

'diversity': 0.8526315789473684,

'avg_overlap': 0.0144720245746806,



Clustering (r/DeepSeek)

Model:

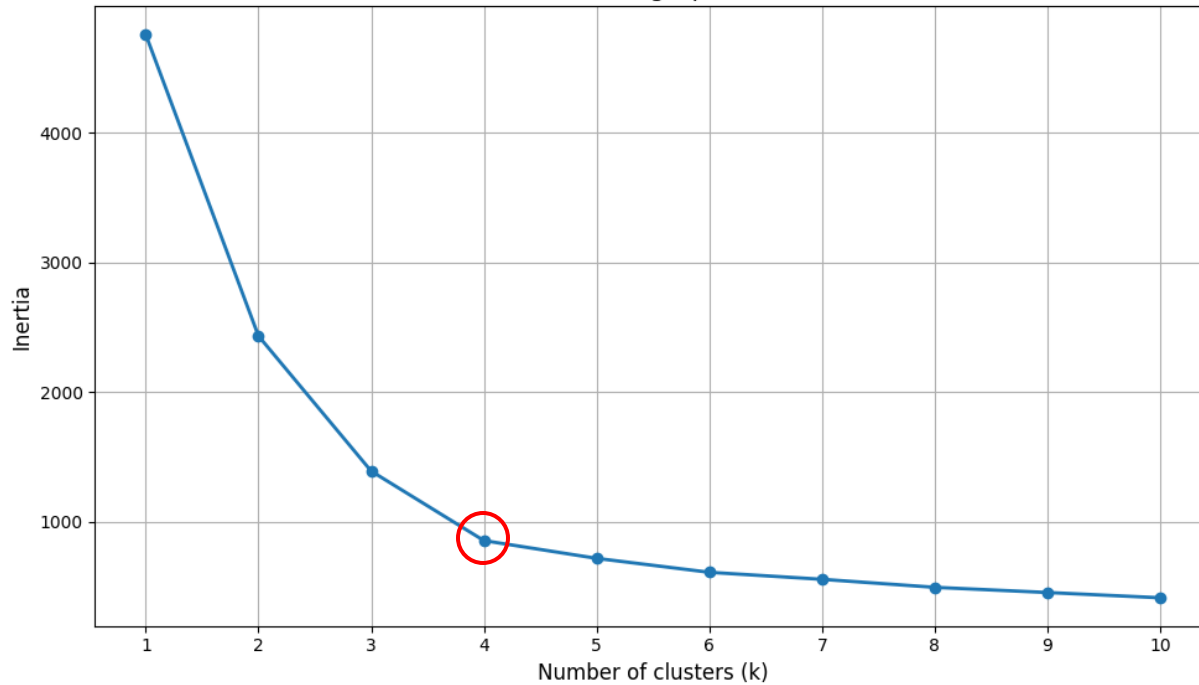
- K-Means applied to sentence embeddings from BERT (e.g., sentence-transformers' all-MiniLM-L6-v2)
- Hierarchical clustering

Process:

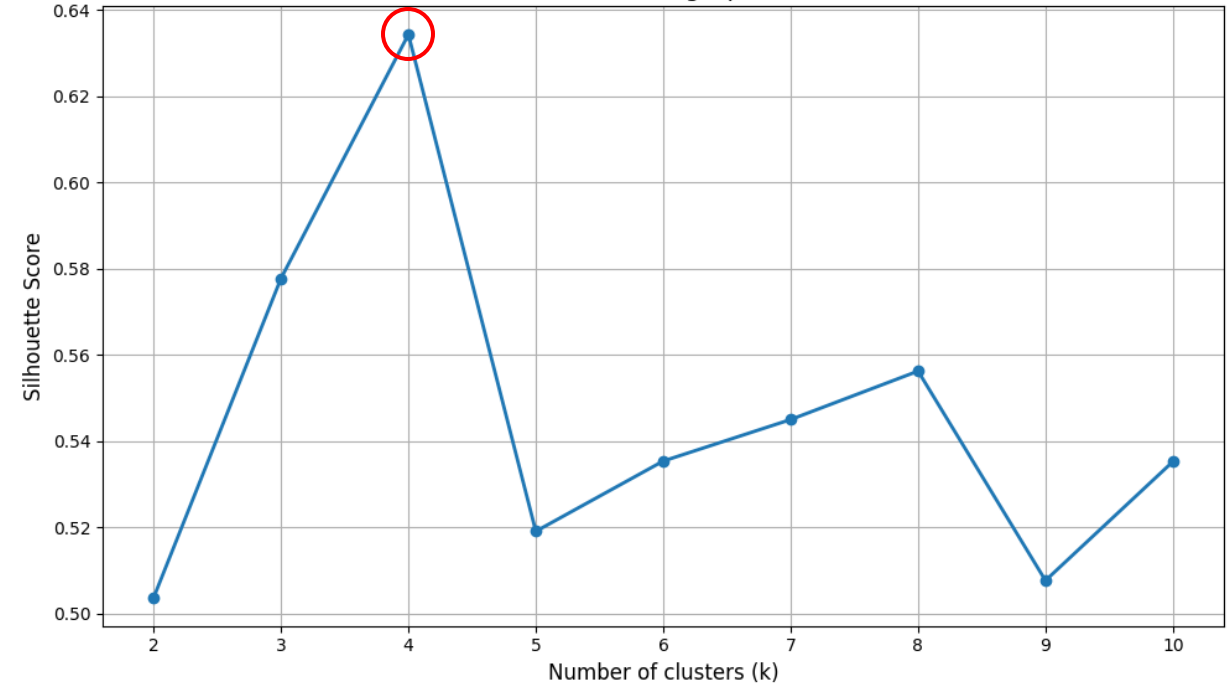
- Generate embeddings for all posts.
- Determine optimal k (e.g., 5–10 clusters) using elbow method or silhouette score.
- Cluster posts and label clusters by analyzing top keywords and representative texts (e.g., "geopolitical threat" cluster).

Finding k (k-means)

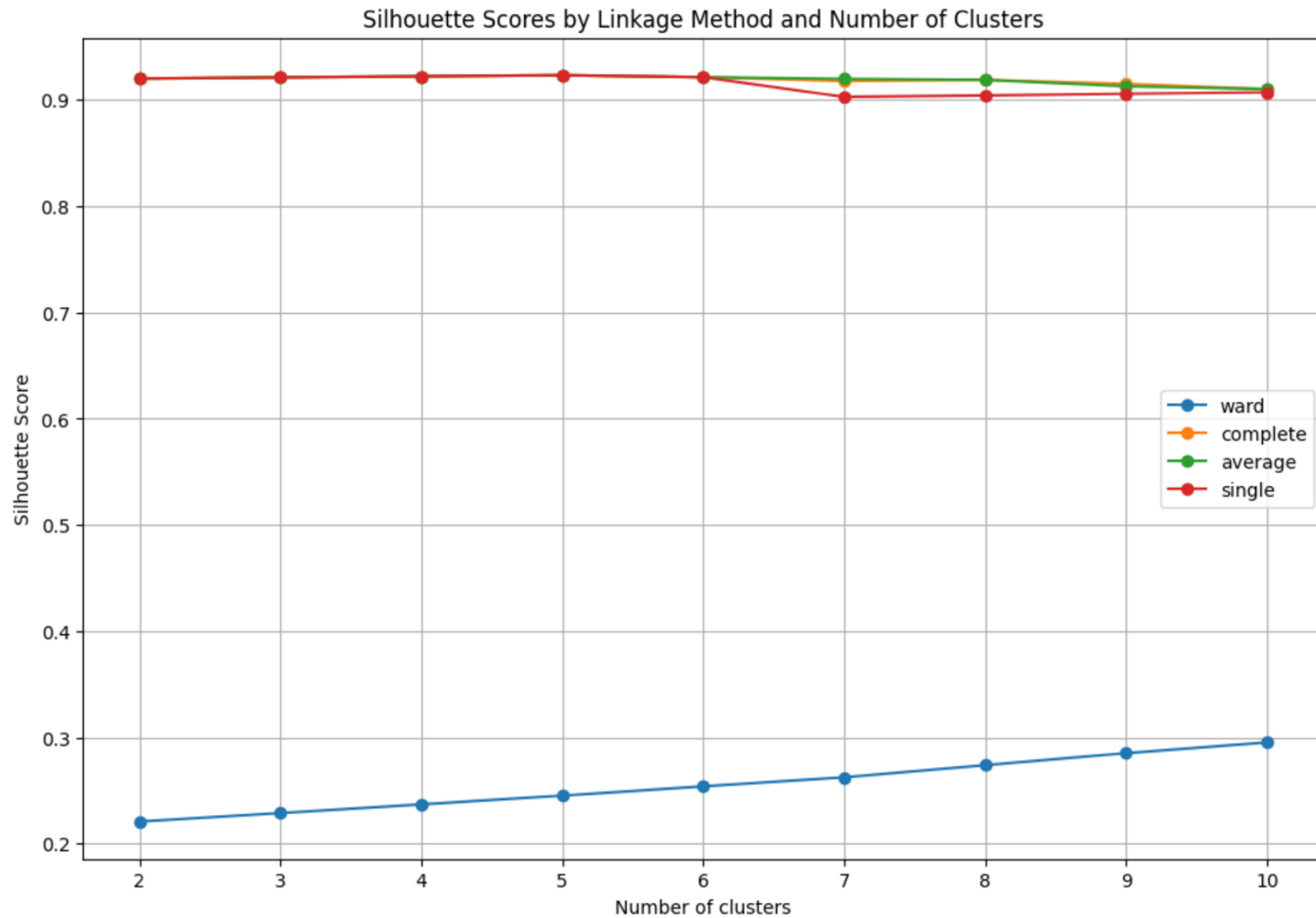
Elbow Method for Determining Optimal Number of Clusters



Silhouette Method for Determining Optimal Number of Clusters

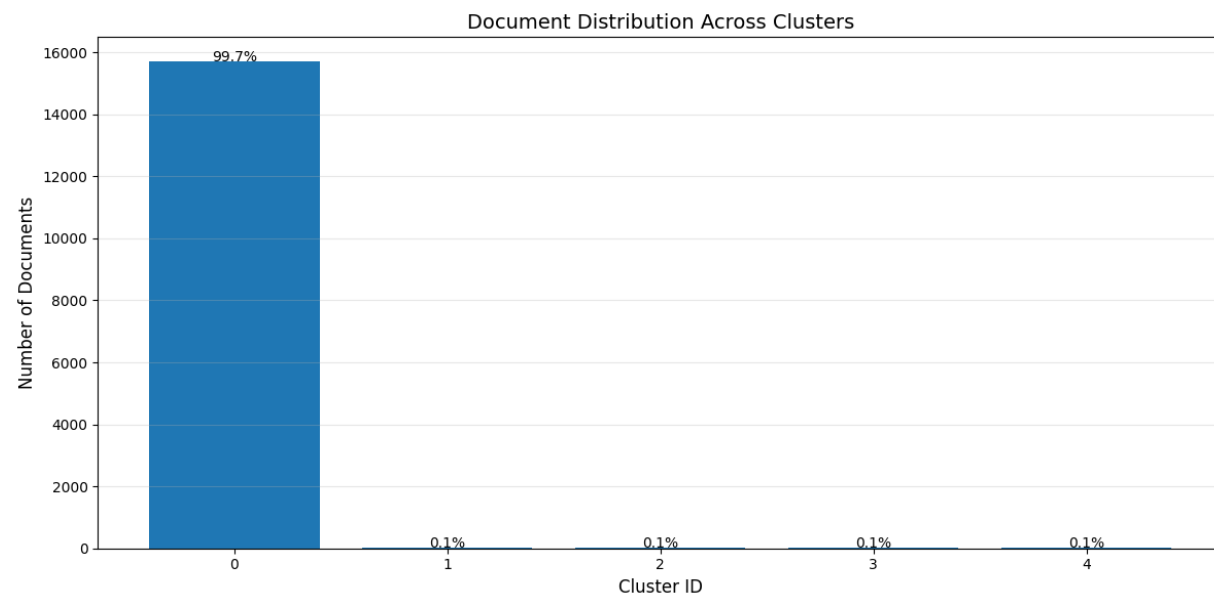
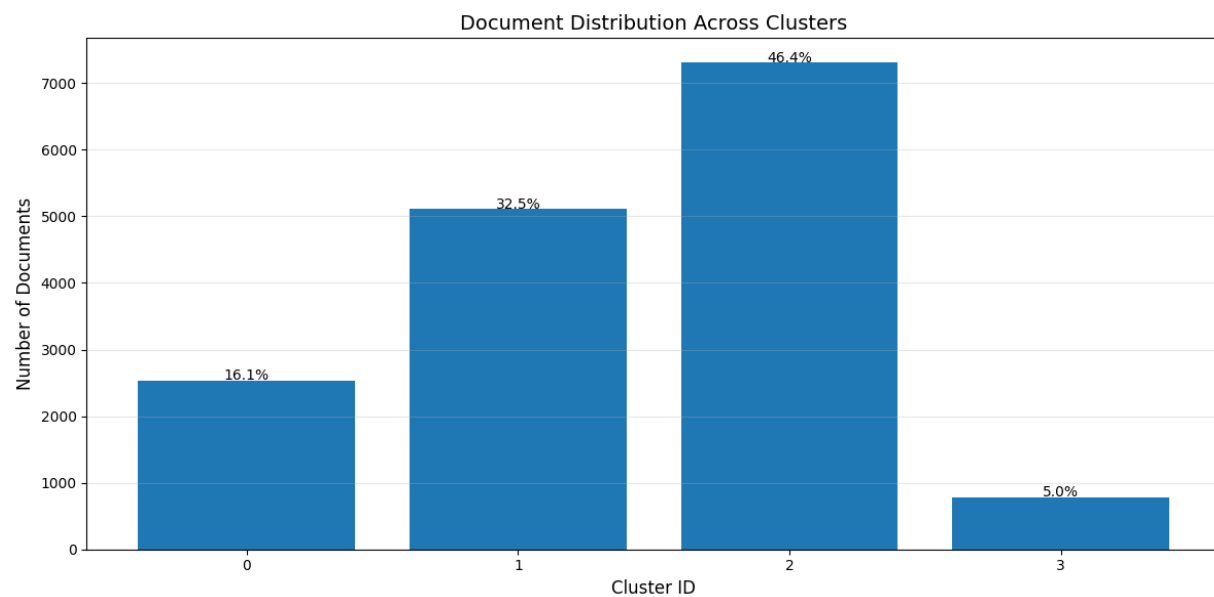


Finding k (hierarchical)



Best linkage method: complete
Best number of clusters: 5

Topic Distribution (k-means vs hierarchical)



Word Cloud (k-means vs hierarchical)



China-US, Propoganda, Censorship, Government



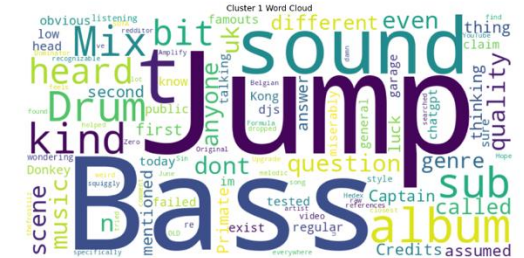
Use, ChatGPT, Server, Time, API



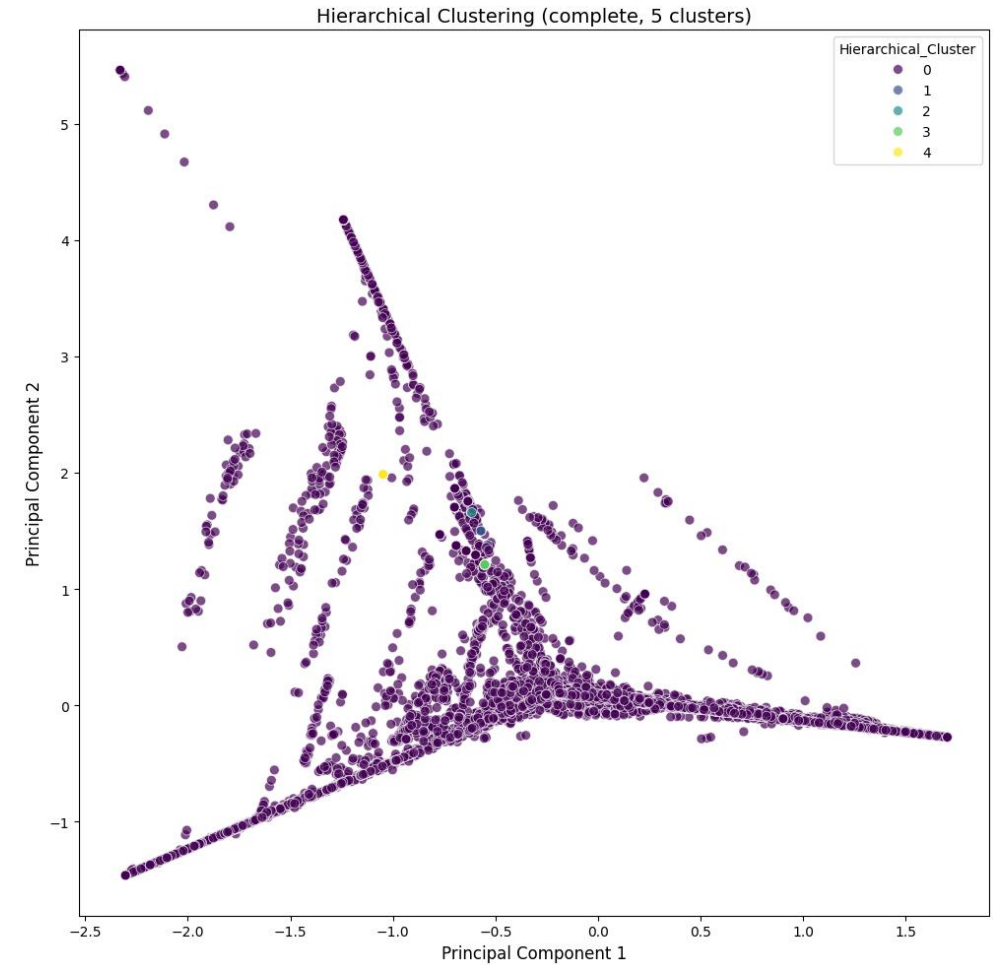
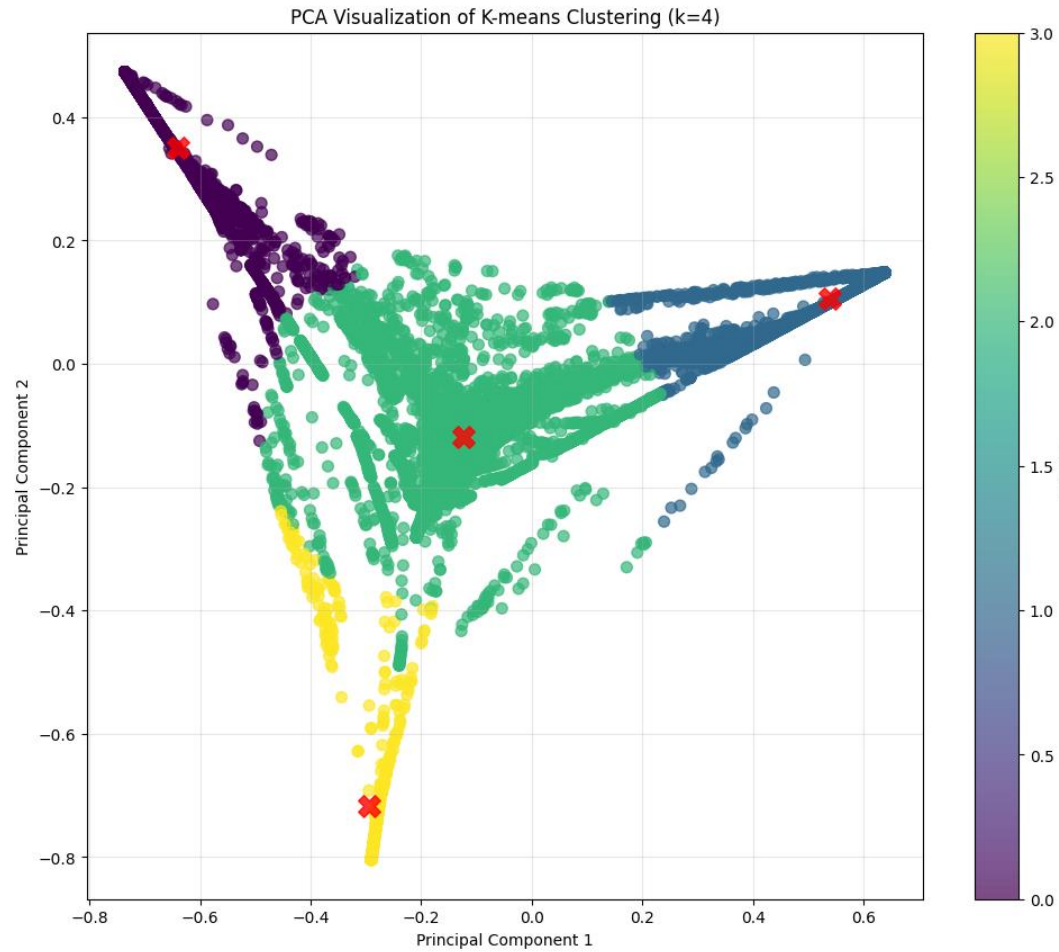
DeepSeek, AI, Model, Use



AI, Use, Prompt

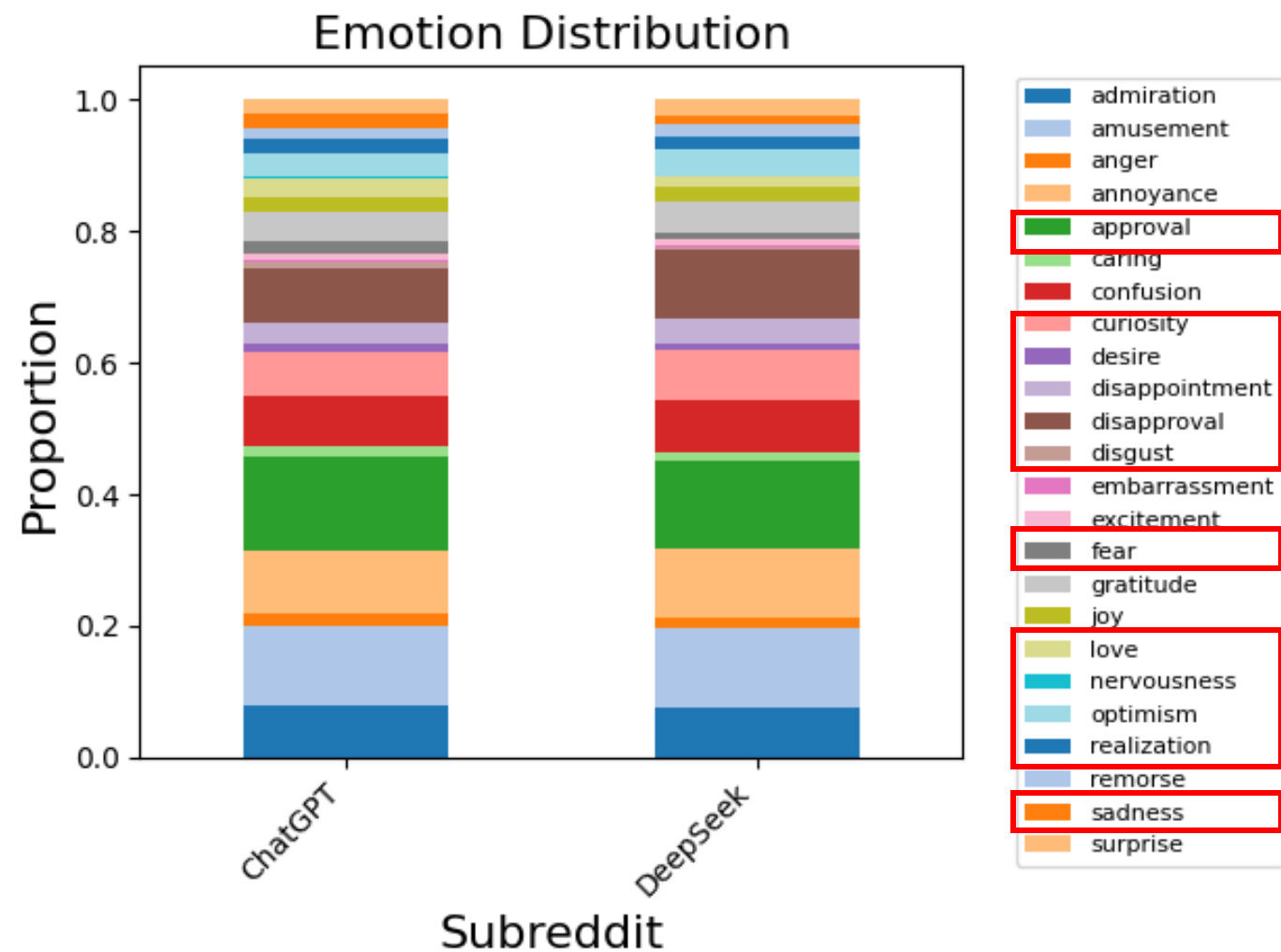


Cluster Overviews (k-means vs hierarchical)

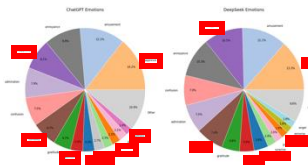


Data Analysis

Emotion Distribution

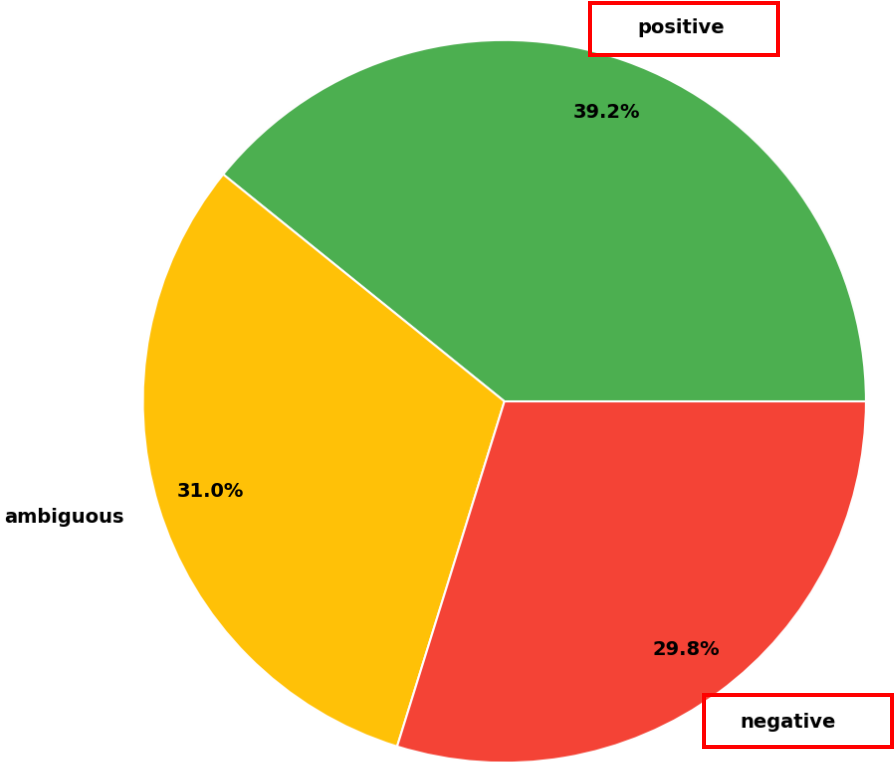


* **Note:** Emotions with red box on legend are different between r/ChatGPT and r/Deepseek statistically significant

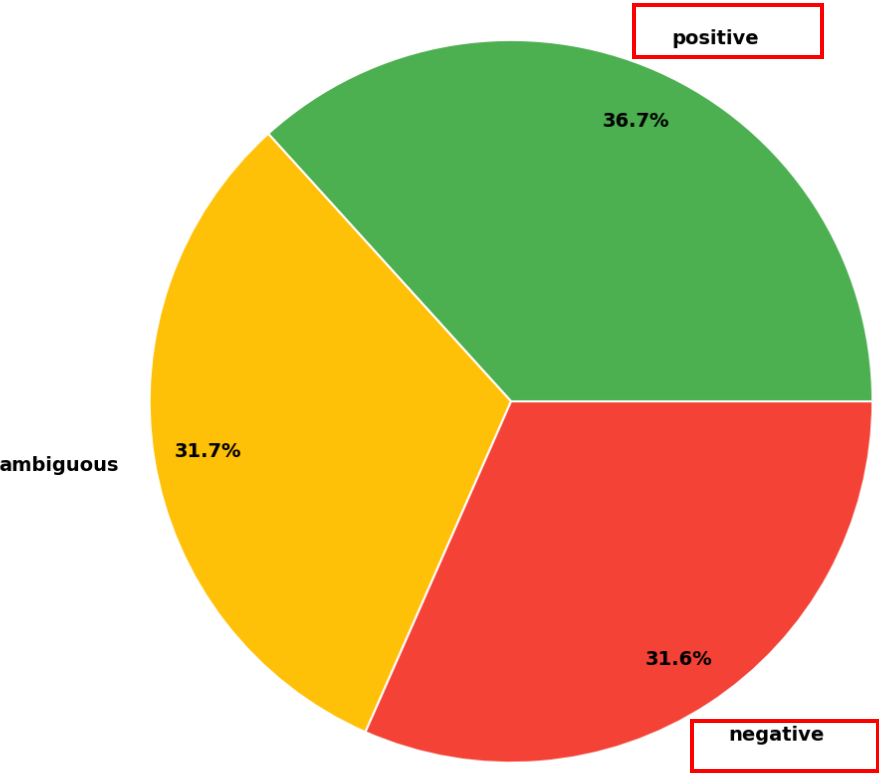


Sentiment Distribution

ChatGPT - Sentiment Categories

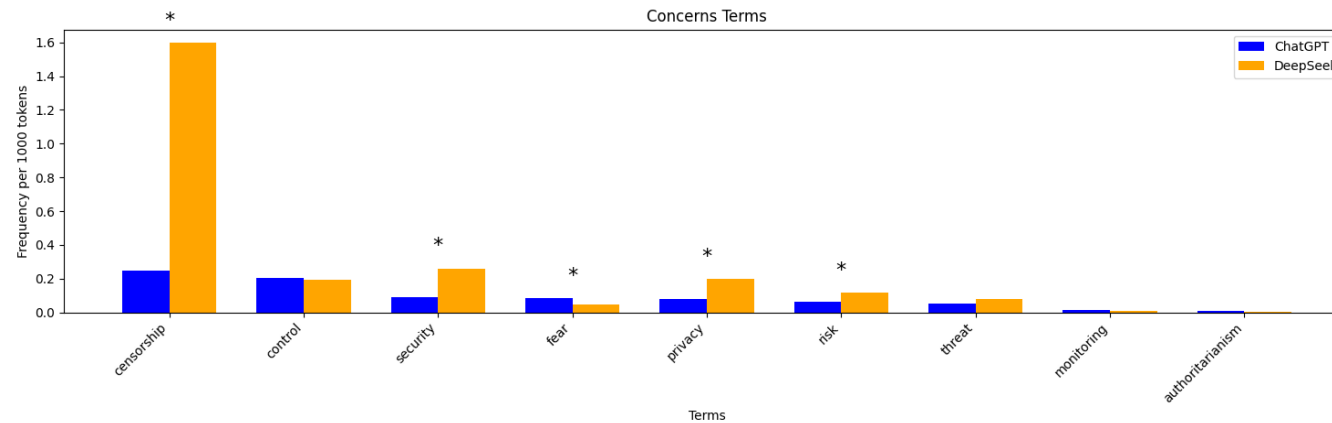
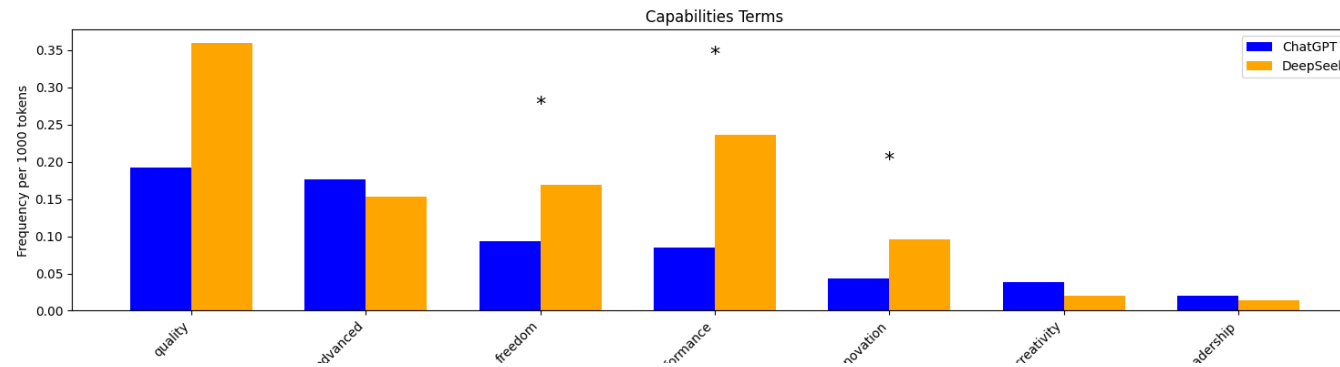
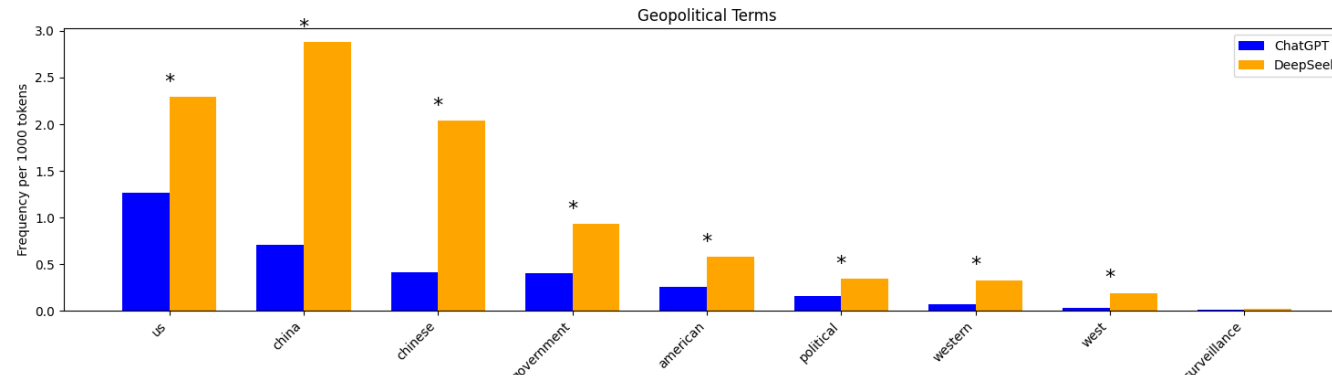


DeepSeek - Sentiment Categories



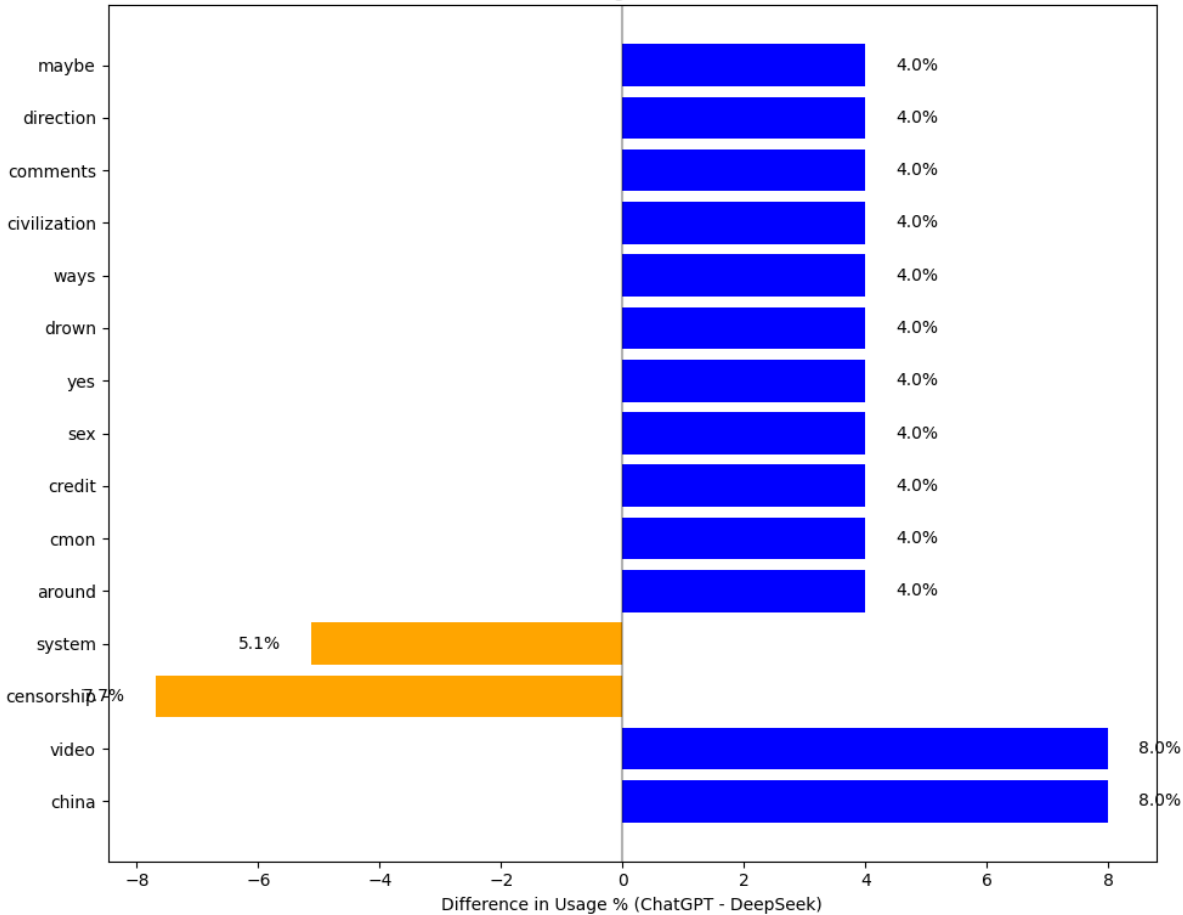
```
Sentiment: negative
{'chi2': 9.033476394352363, 'p_value': 0.0026507989413307215, 'is_significant': True}
{'z-stat': -3.019824603192709, 'p_value': 0.0025292110778380953, 'is_significant': True}
Sentiment: positive
{'chi2': 14.893010644253074, 'p_value': 0.00011378378002645242, 'is_significant': True}
{'z-stat': 3.872527356323686, 'p_value': 0.00010771258170140623, 'is_significant': True}
```

Terminology Usage

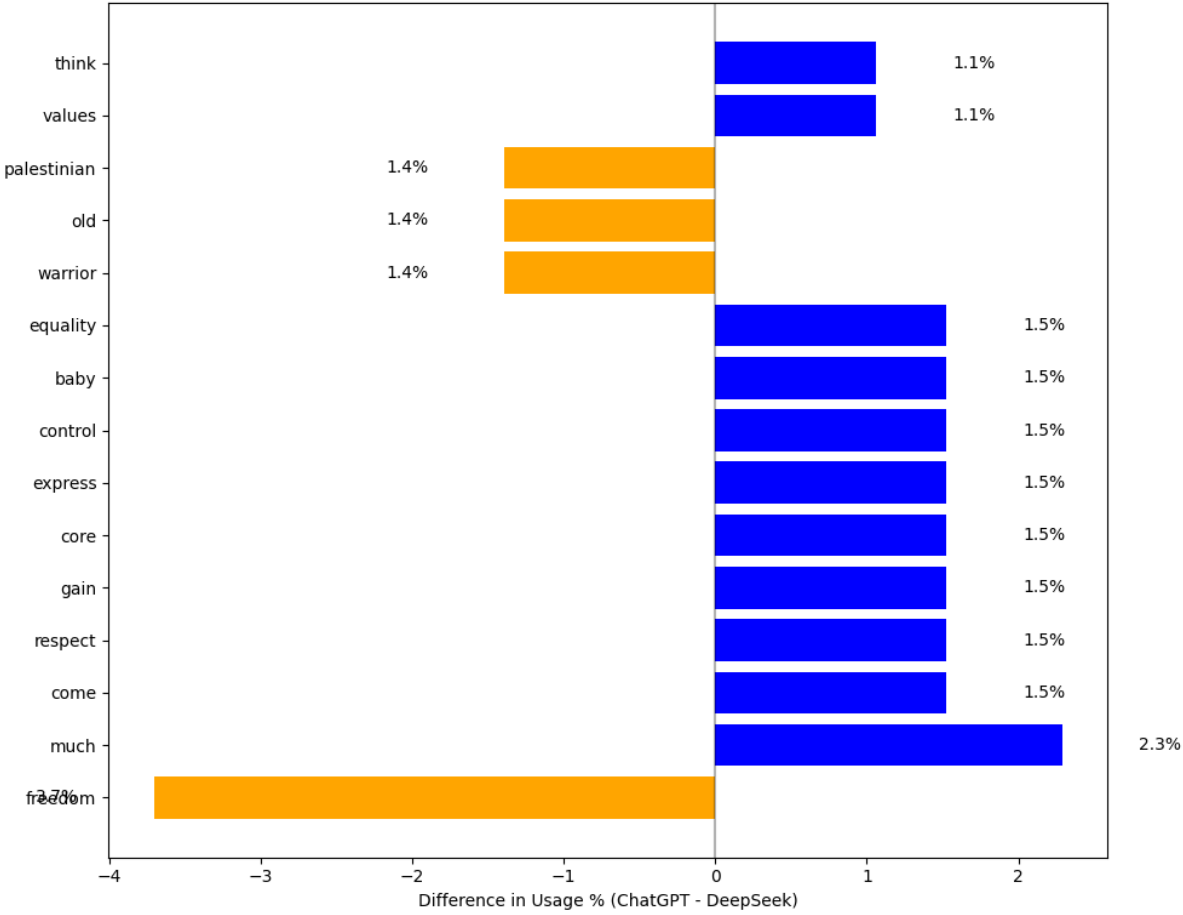


Word Analysis

Contextual Word Usage Around "surveillance"



Contextual Word Usage Around "freedom"



[illegible][illegible][illegible][illegible]

UI/UX, prompt engineering

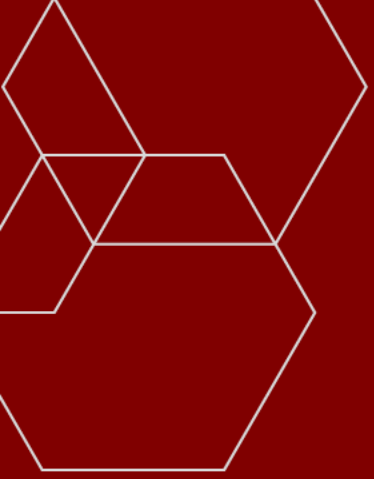
Discussions

- **Limitations:**

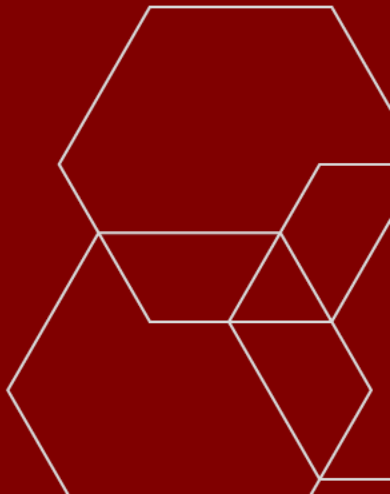
- Reddit sample represents tech-savvy Western users, not general population
- Limited time frame (3 months) may not capture long-term trends
- Clustering may oversimplify nuanced discussions

- **Outlook:**

- Comparative analysis with more AI models like Claude, Grok, Gemini, Qwen, etc.
- Improved temporal modeling to capture impact of external events
- Visual content and meme analysis could enrich understanding
- Comparative analysis with Chinese-language discussions would provide balance
- Clustering framework could support recommendation systems to diversify content exposure



THANK YOU!



References

Papers:

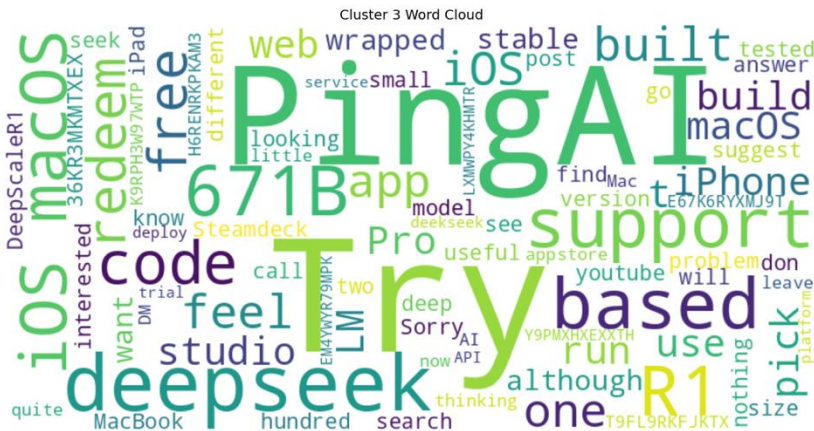
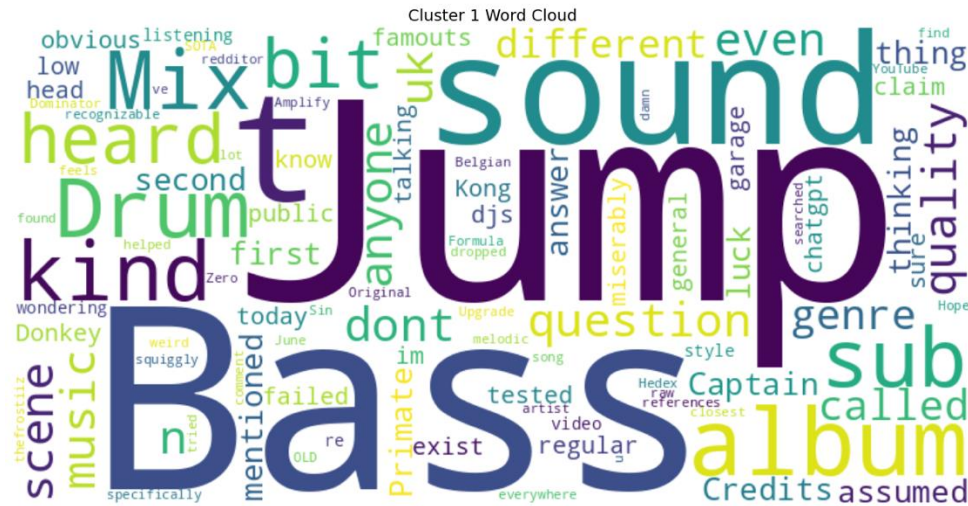
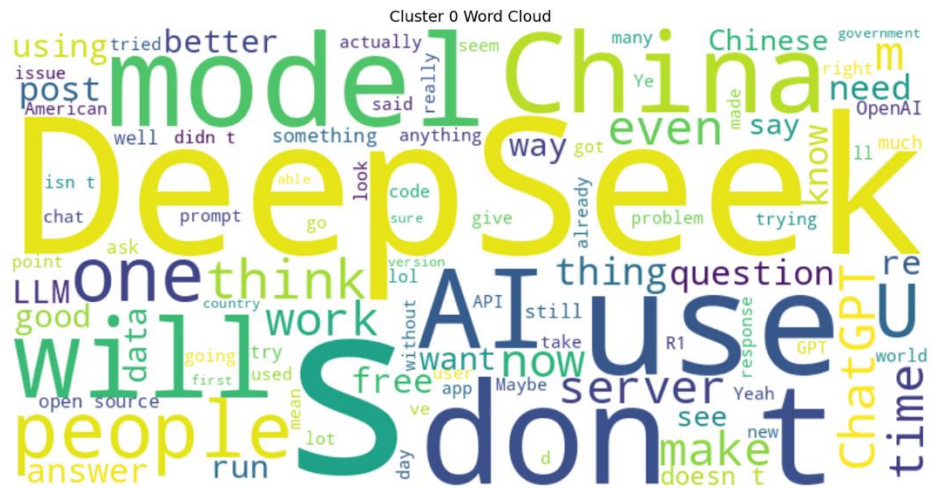
- Fast, E., & Horvitz, E. (2017). Long-Term Trends in the Public Perception of Artificial Intelligence. *Proceedings of the AAAI Conference on Artificial Intelligence*, 31(1).
<https://doi.org/10.1609/aaai.v31i1.10635>
- Said, E. W. (1977). Orientalism. *The Georgia Review*, 31(1), 162-206.
- Said, E. W. (2023). Orientalism. In *Social theory re-wired* (pp. 362-374). Routledge.

Models:

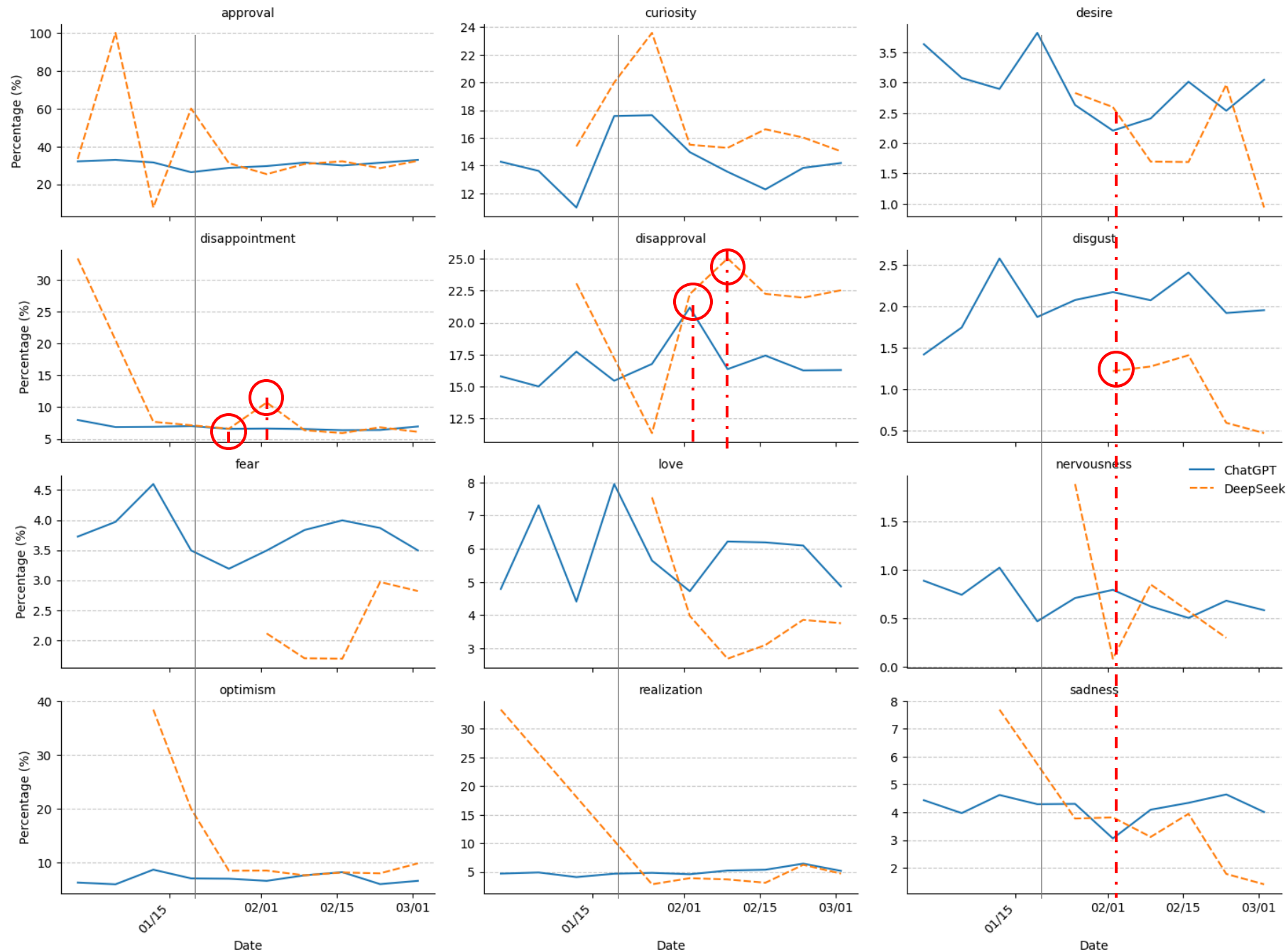
- https://huggingface.co/SamLowe/roberta-base-go_emotions
- <https://maartengr.github.io/BERTopic/index.html>

Appendix

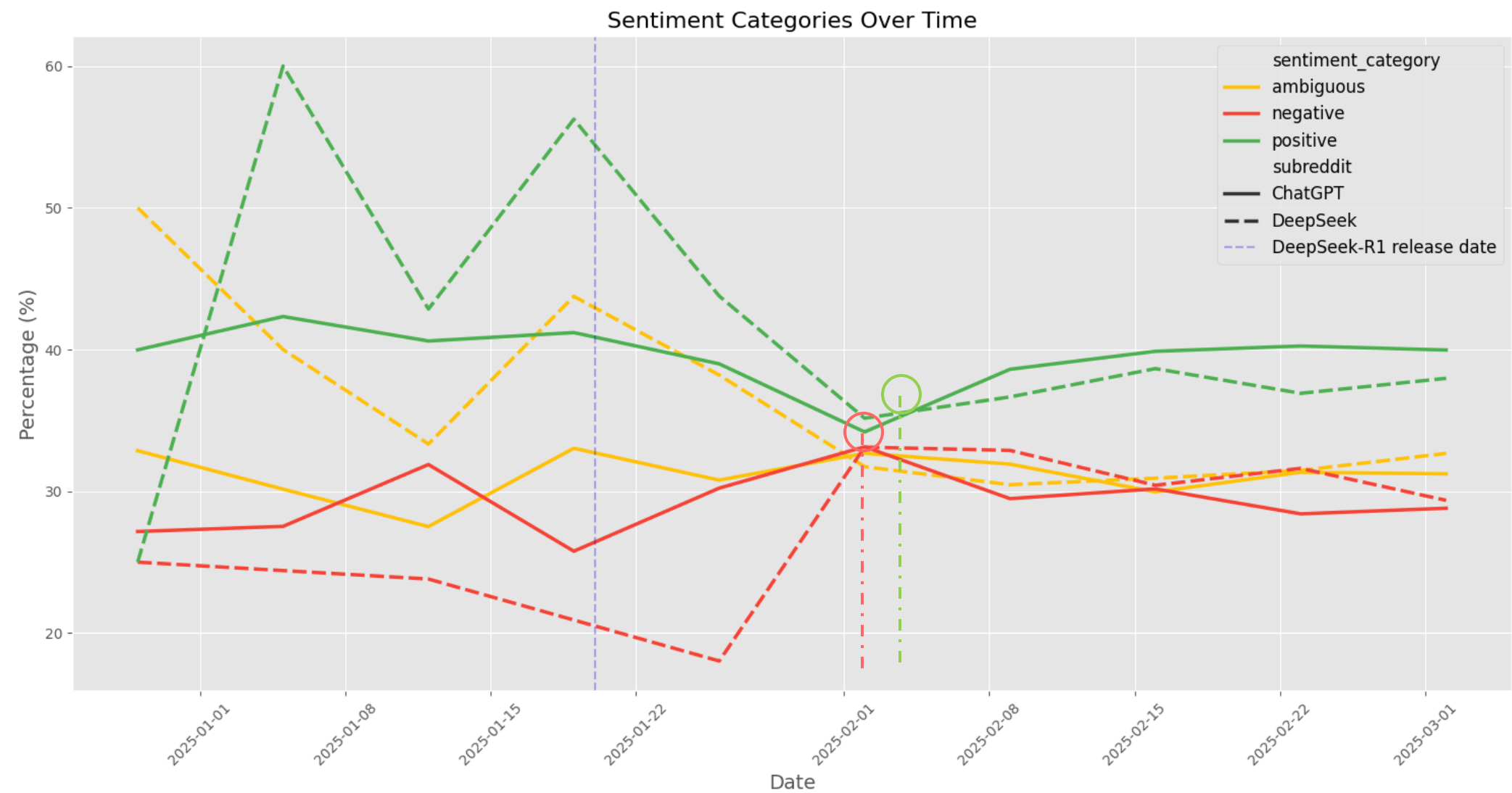
Word Distribution (hierarchical clustering)



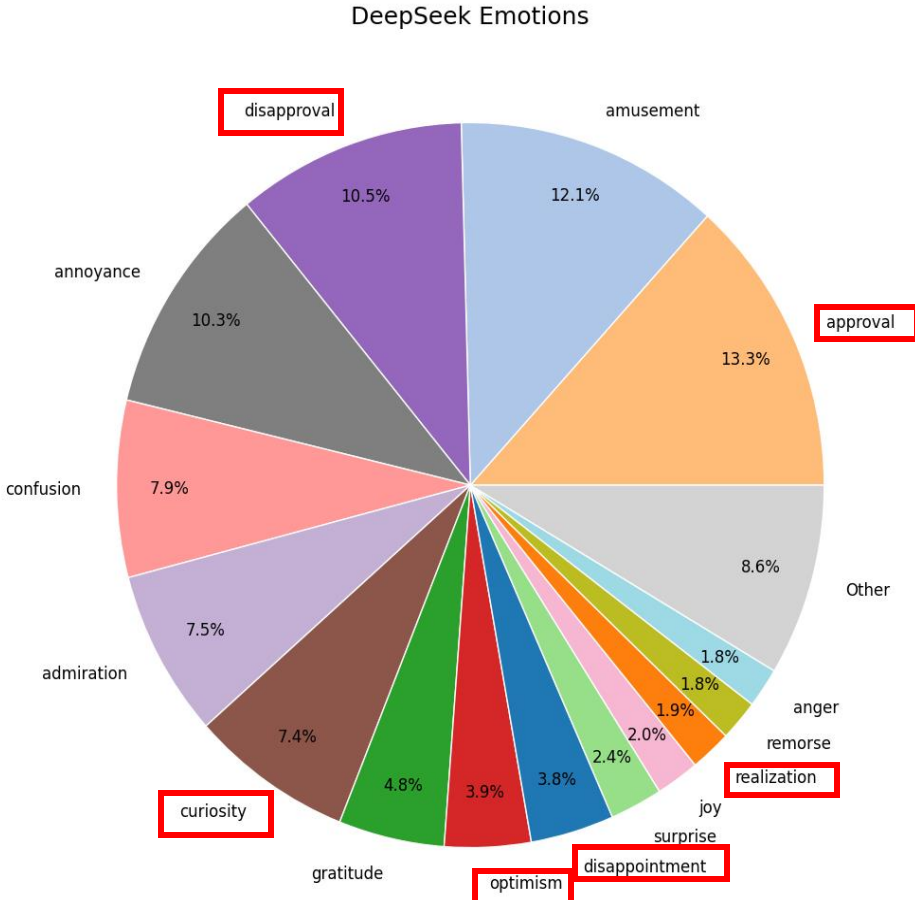
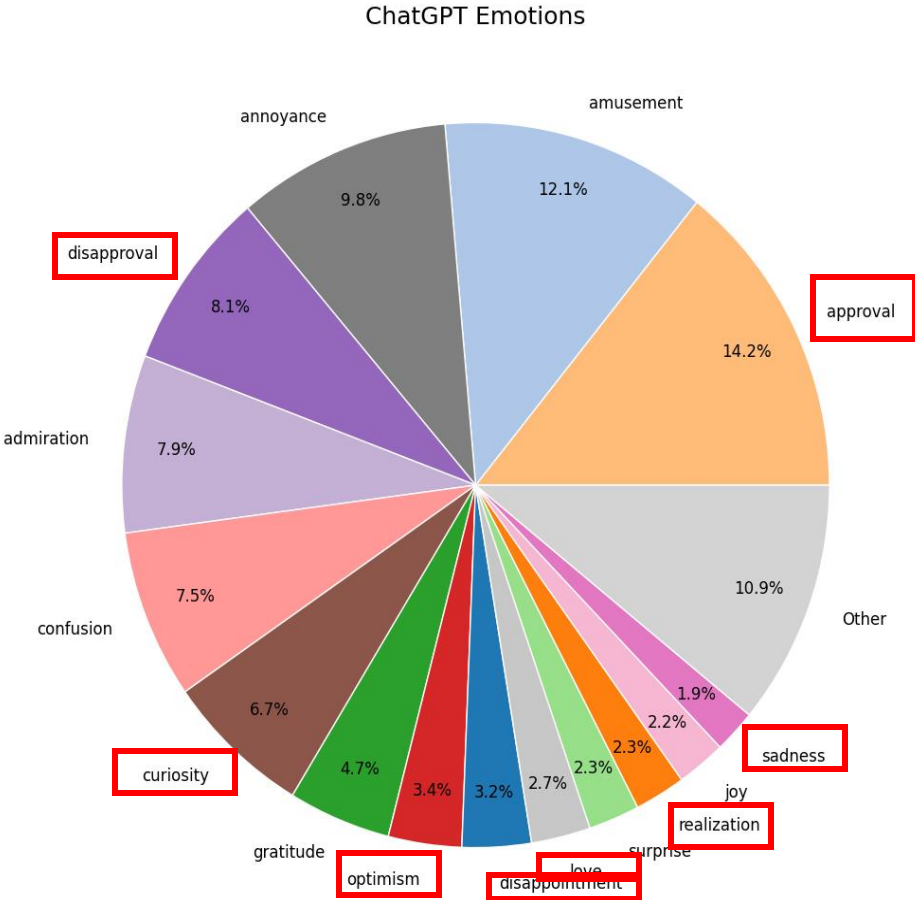
Temporal Analysis Of Emotions



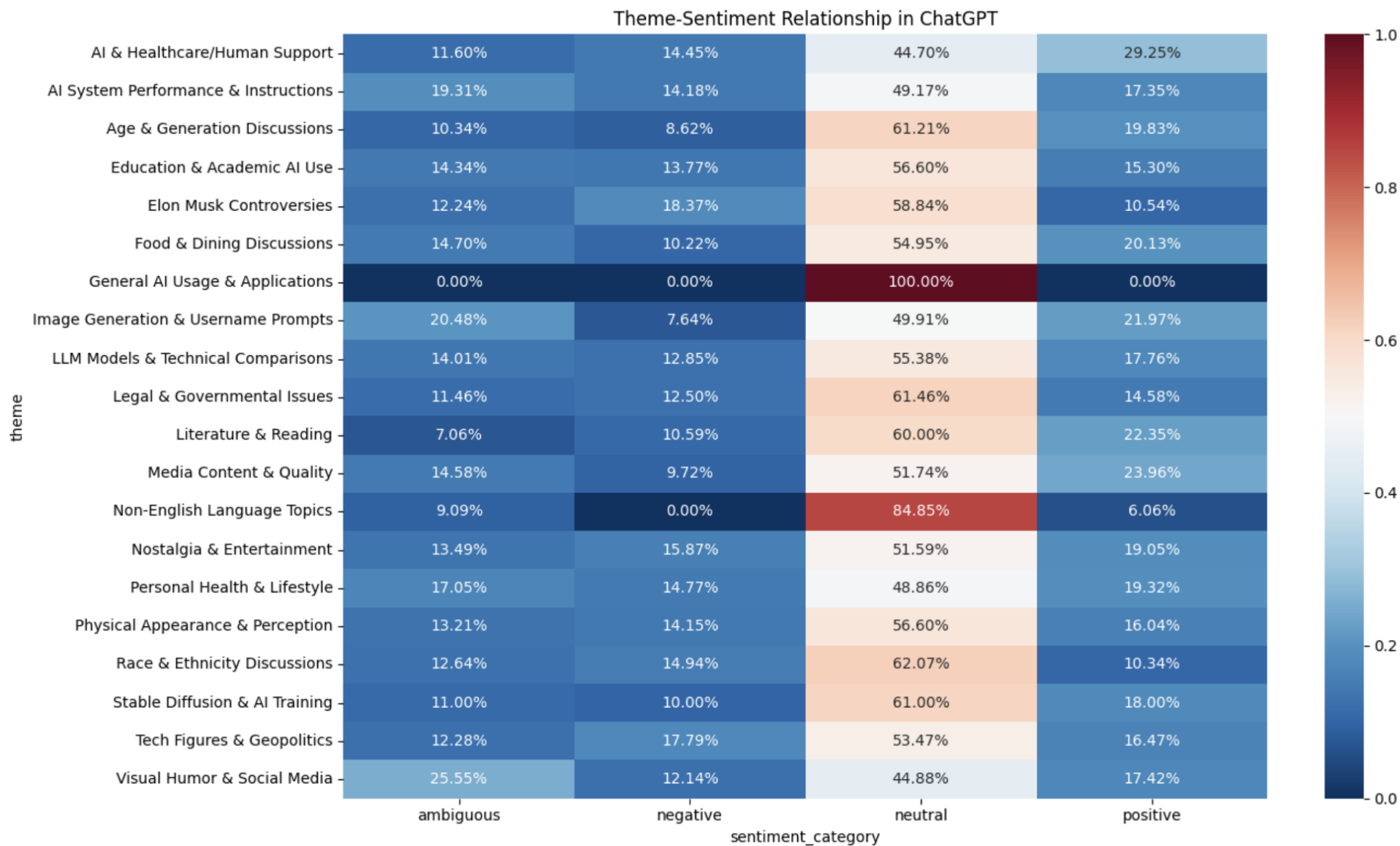
Temporal Analysis of Sentiment



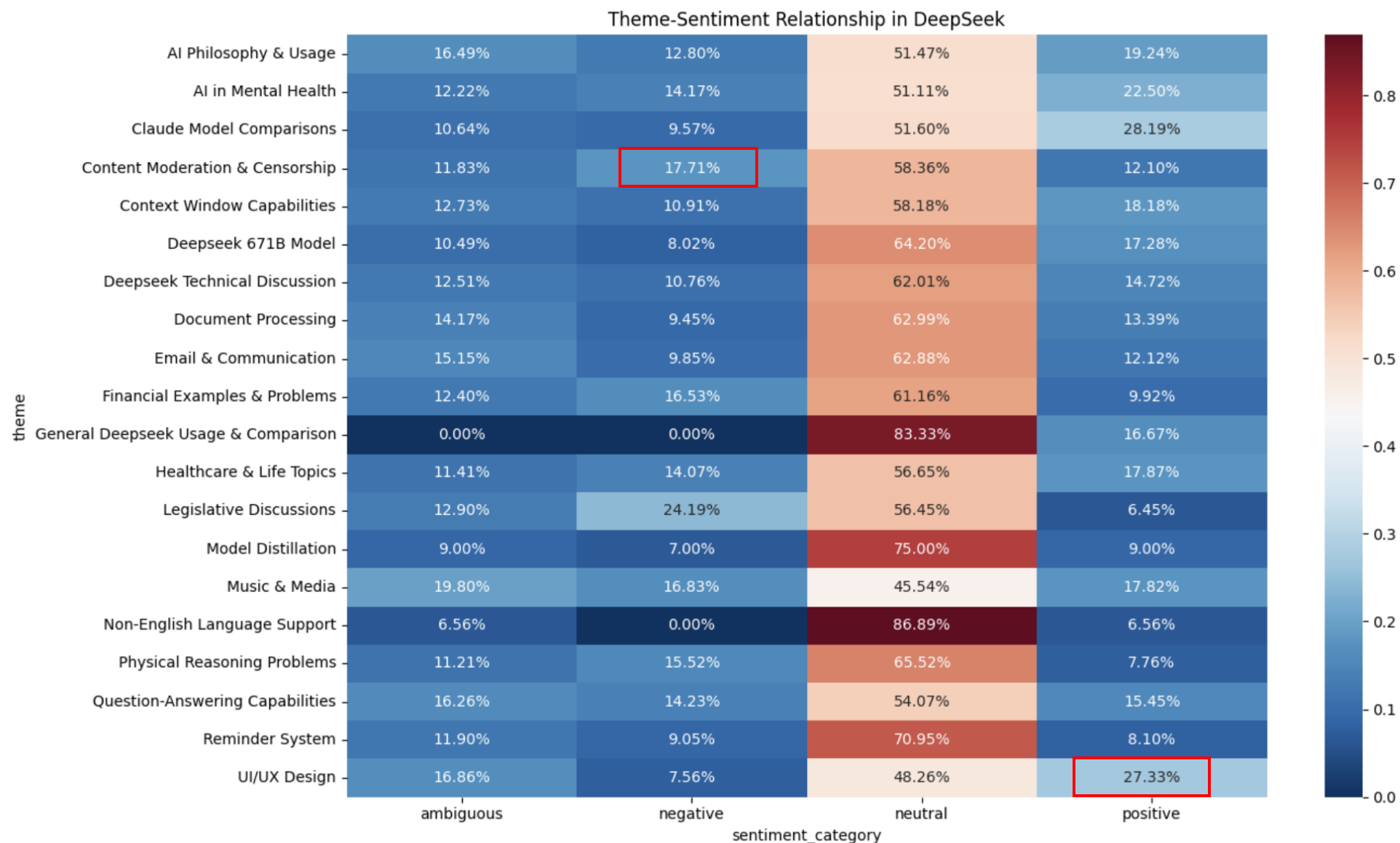
Emotions Distribution



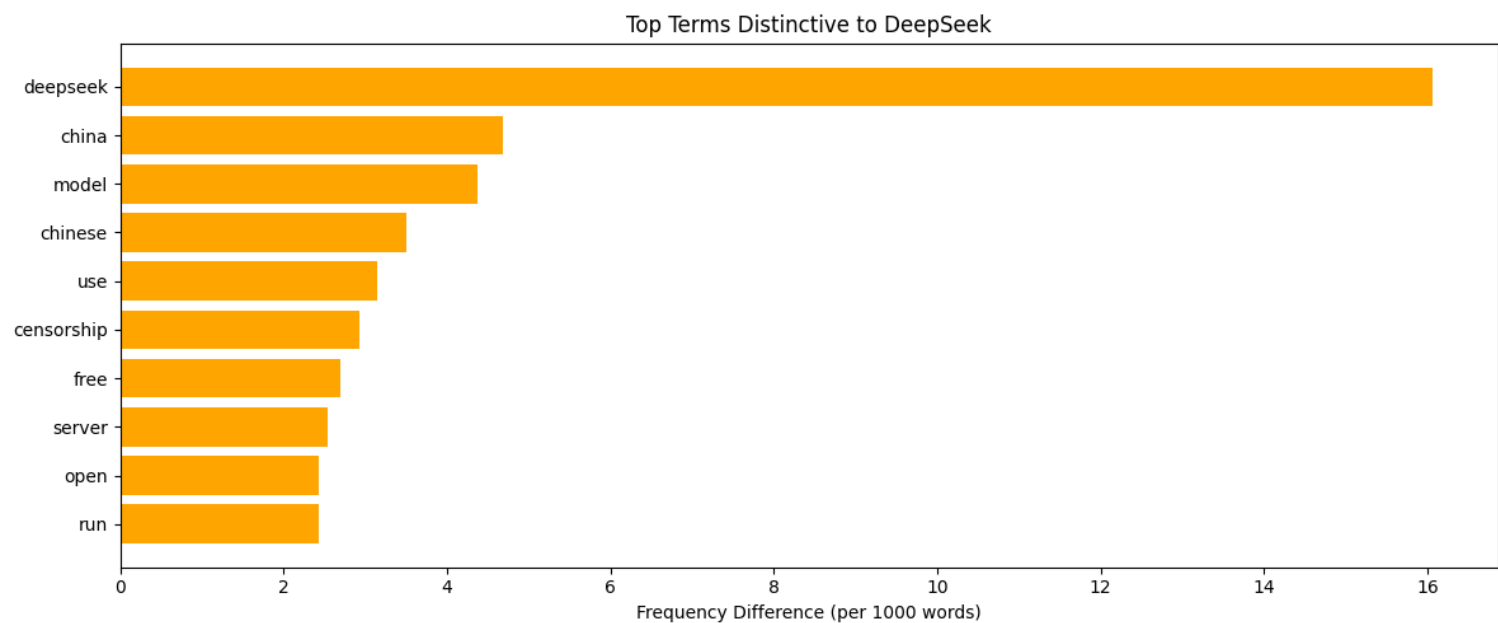
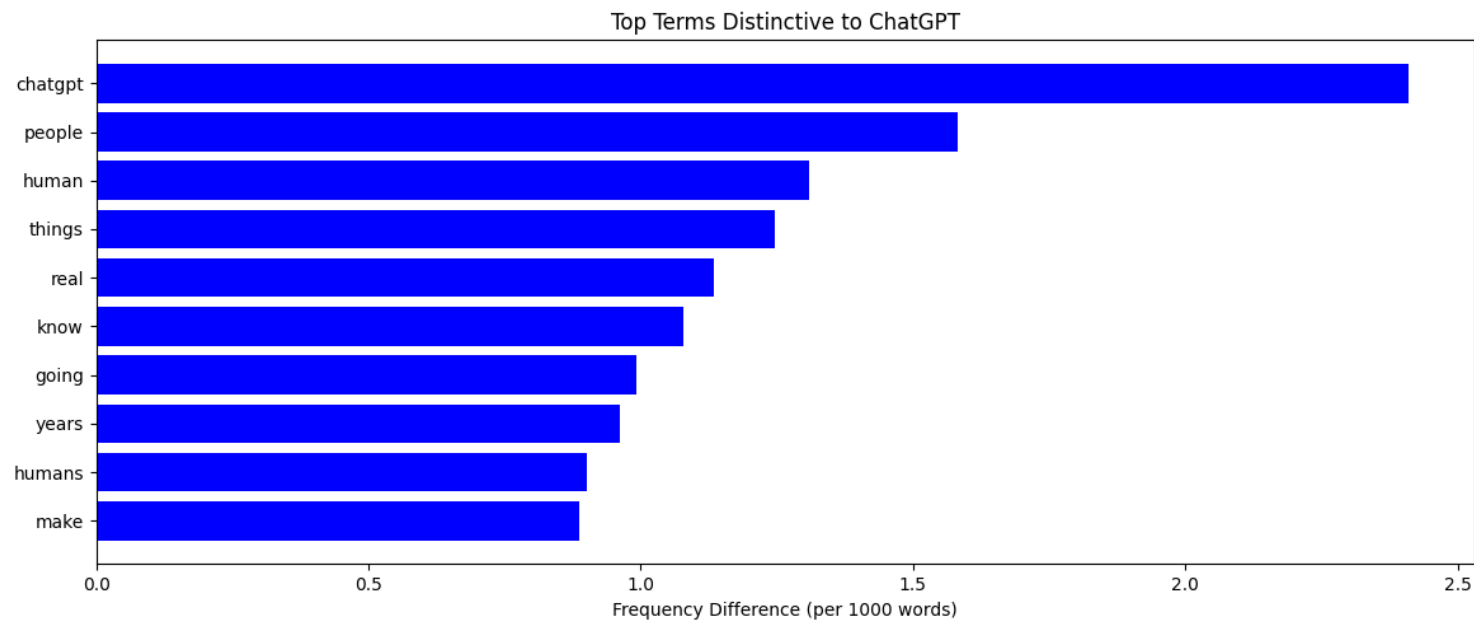
Theme-Sentiment Relationship of ChatGPT



Theme-Sentiment Relationship of DeepSeek



Top Terms Distribution



Contextual Word Usage of 'ChatGPT' and 'DeepSeek'

